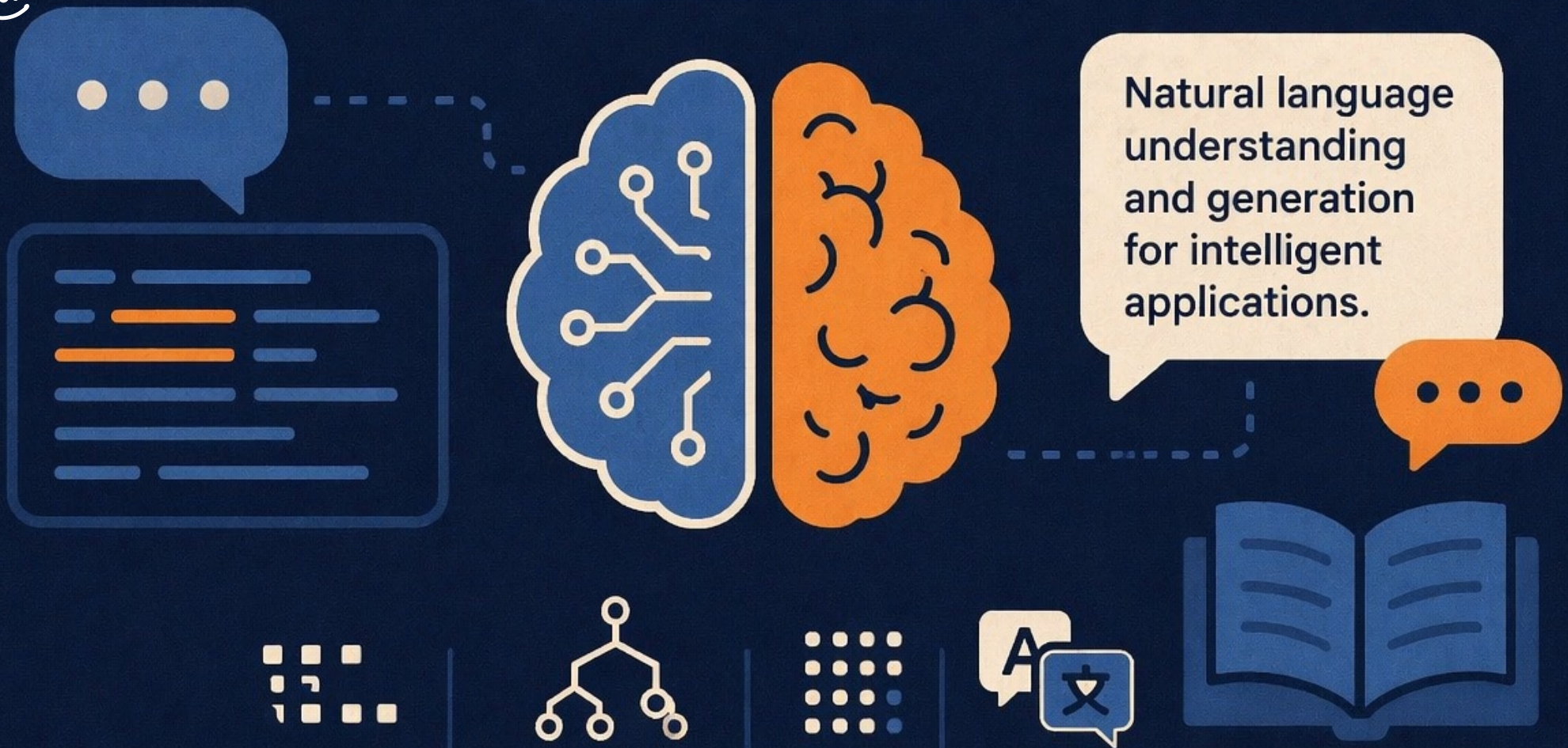
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: Word Embeddings, Sentiment Analysis and Topic Modelling

#Agenda

1) Word2Vec Recap
2) Word2Vec Code
3) Sentiment Analysis
4) Topic Modelling

Recap

text = "I love NLP", "Word2Vec is so cool"

→ pre-processing (lower, removing stopwords, punct, stem/lemma, link Numbers)

→ Break it into tokens (Converting text to some Numbers)

I ⟶ [1234] ⟶ token id

love ⟶ [8625]

NLP ⟶ [5843]

Just with tokenization, we cant get symantic meaning

Happy → [6823]

Q → Joy → [7013]

↳ query

"Word Embedding"

Happy → Vector rep [- - - - -]

Embedding

"close by"

Joy → [- - - -]

Embedding

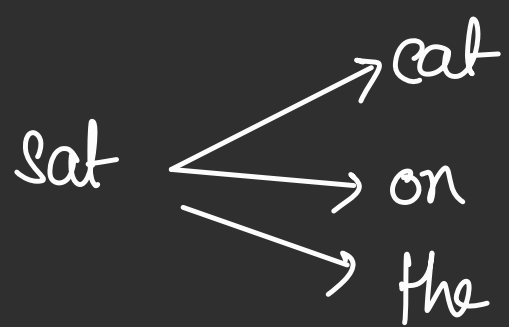
Word2Vec

1) CBOW → Fill in the blanks
   - context word → I/P
   - Target word → O/P
2) Skip-gram
   - Target word → I/P
   - Context word → O/P

Neural N/W

Shallow (1 hidden layer)

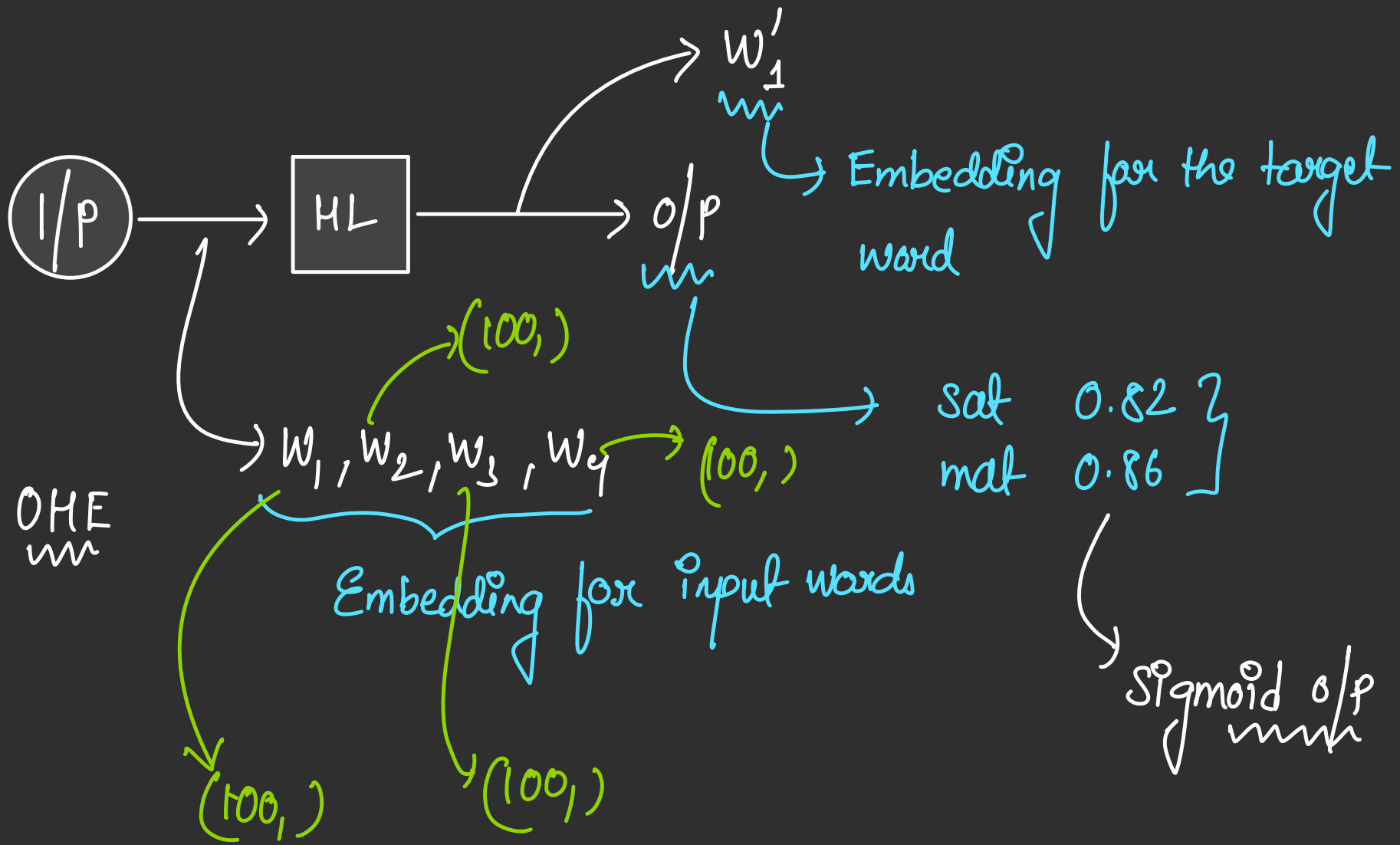
Weights of NN are the embeddings



✓ ✓ ✓ ✓
cat __ on the mat

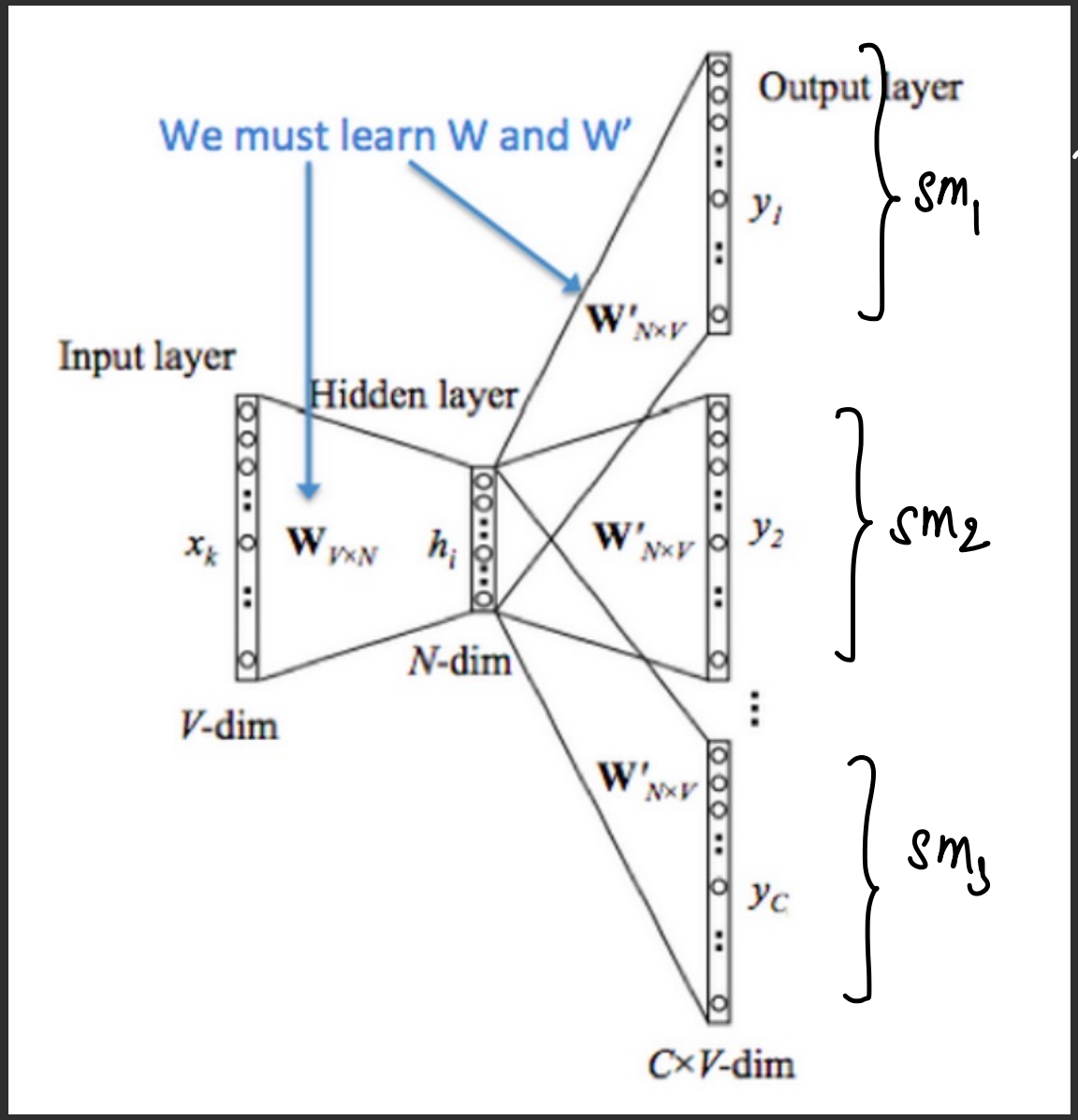
__ → Target

CBOW


I/P
HL
O/P
W'_1
Embedding for the target word
cat
on
the
mat
OHE
W_1, W_2, W_3, W_4
(100,)
(100,)
(100,)
(100,)
Embedding for input words
sat 0.82
mat 0.86
Sigmoid o/p

#Negative sampling

$$\frac{e^{z}}{\sum_{i=1}^{n} e^{z_i}}$$



→ denominator

all the words in corpus

count multi-class

↓

Binary class

pos sample pair

(cat, sat)

Neg sample pair

(on, the)

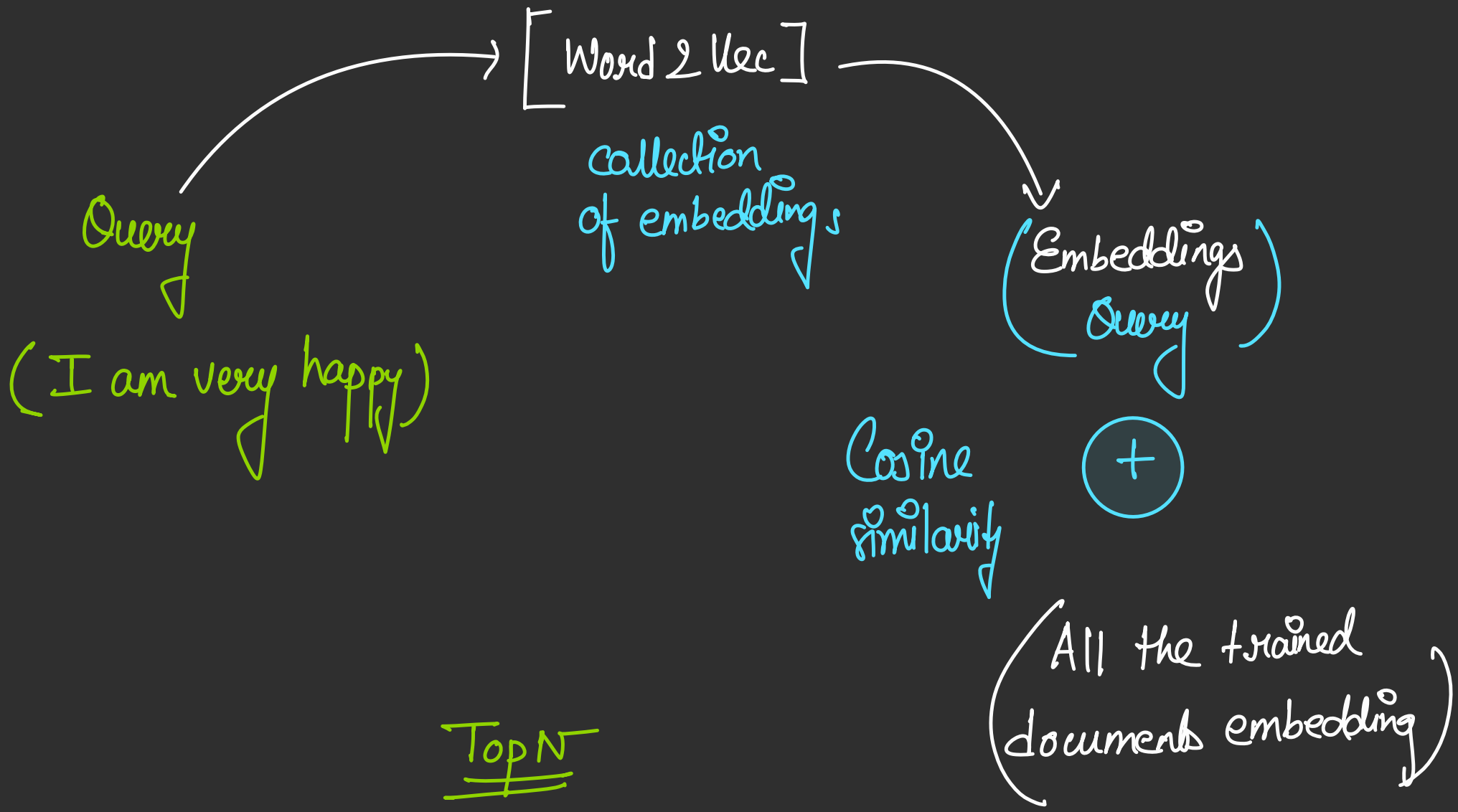
(cat, my)

(stable, could)

✓ ✓

cat sat on

↳ I/p


Word 2 Vec
collection of embeddings
Query
(I am very happy)
Embeddings
Query
Cosine similarity
+
All the trained documents embedding
Top N

problems with Word2Vec

1) No symantic relation across same word.

Apple Apple

Fruit Company

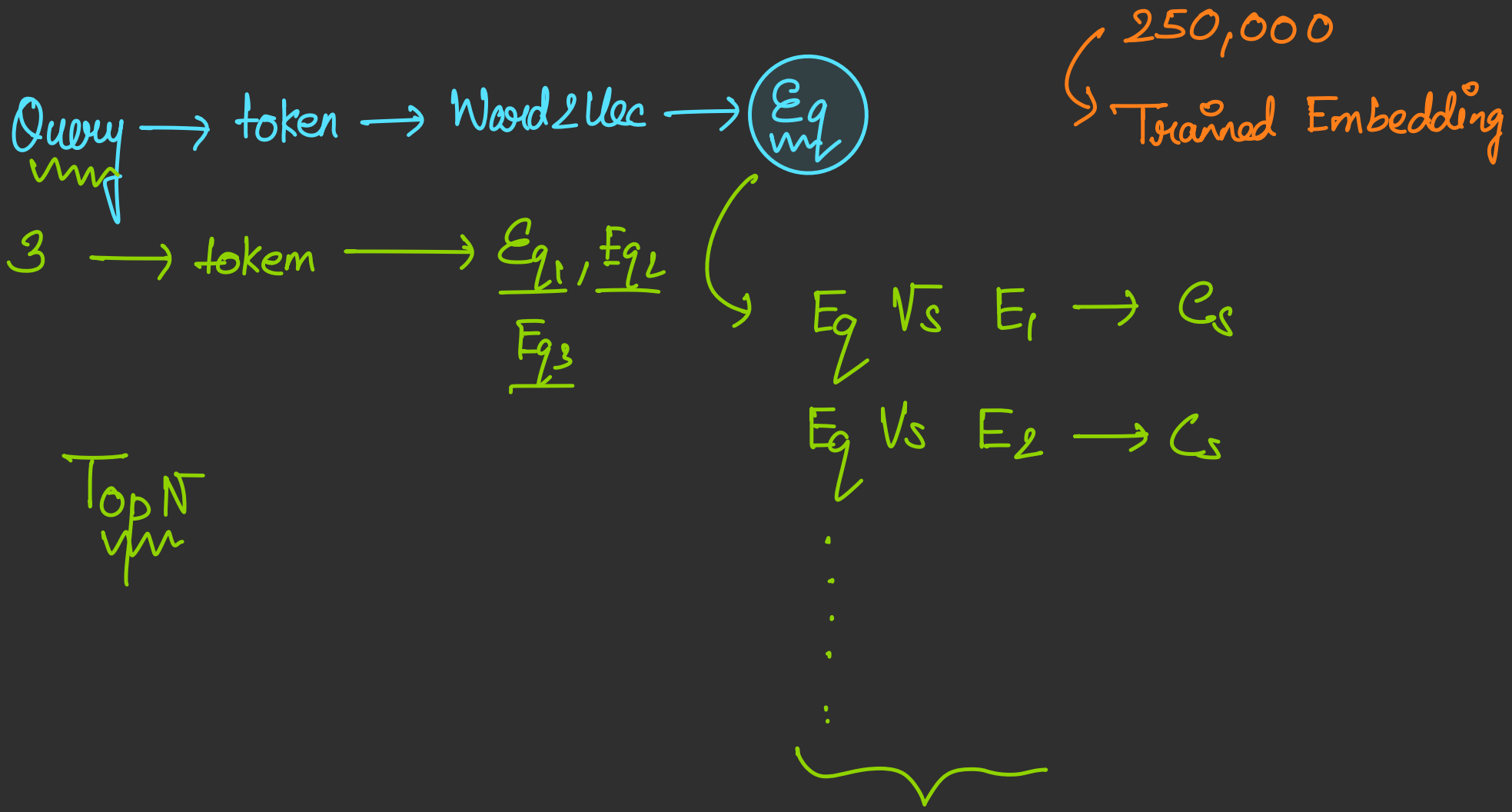
3) Training the Network

2) Embedding size

100 Vs 256

128

Vector DB


Query → token → Word2Vec → Eq
250,000
Trained Embedding
3 → token → Eq1, Eq2
Eq3
Eq Vs E1 → Cs
Eq Vs E2 → Cs
Top N

"Word Embeddings"

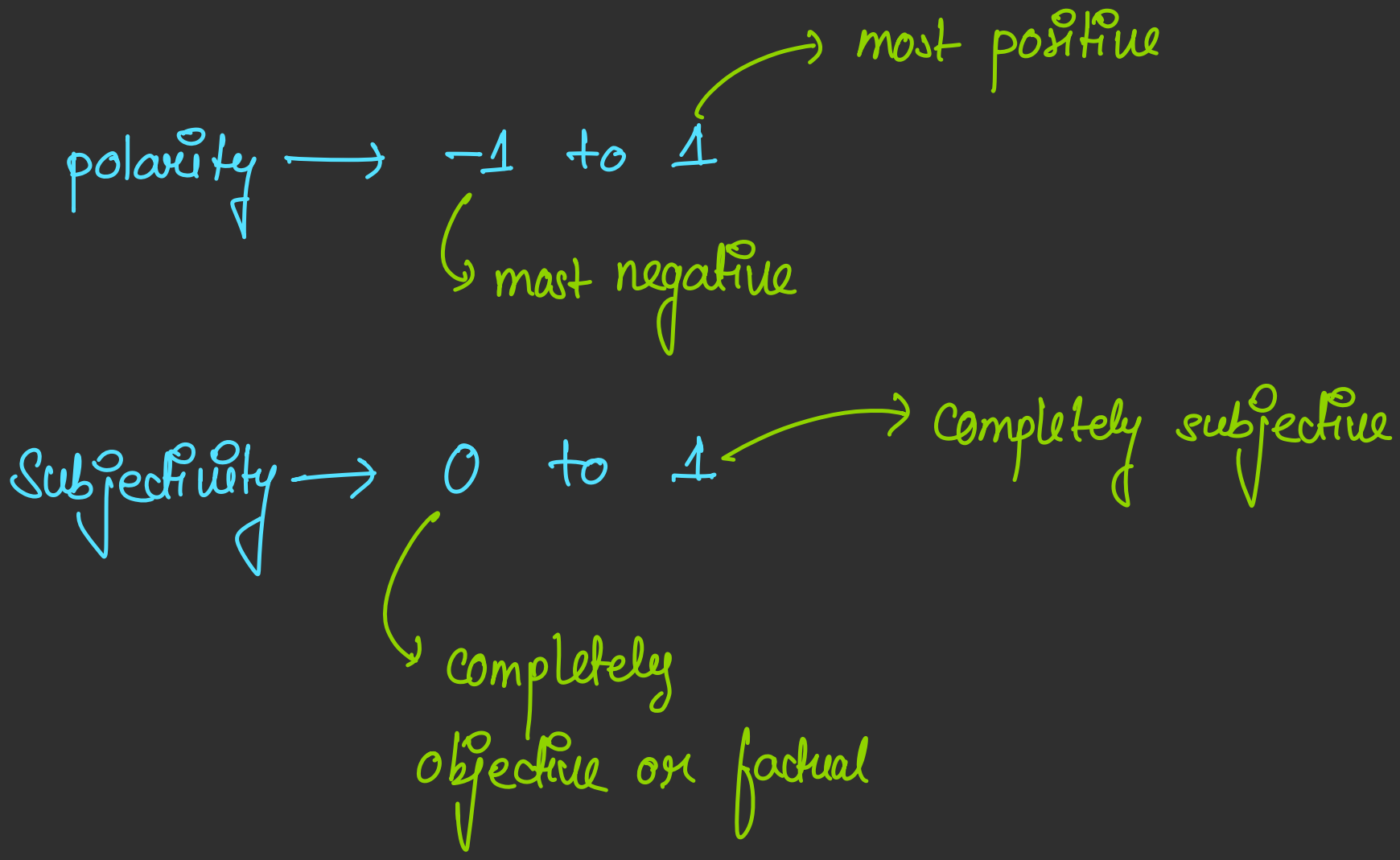
Happy ≈ Joy

Lexicon Based Sentiment Analysis

"The service of that place was great"

↳ pos

#TextBlob


polarity → -1 to 1
most positive
most negative
Subjectivity → 0 to 1
completely subjective
completely objective or factual

Rule based → ANN → RNN → LSTM → "Attention is all you need"

↓

(Transformer)

↓

LLM (GPT, BERT)

(pre-trained)

gpt-4 → 20GB

20GB SSD

(I am hungry)

E_1 E_2 E_2

Doc2Vec

↳ Sentence Embeddings

Mean (E_1, E_2, E_3)

{ Open AI Embeddings
  Sentencence BERT

↳ Sentence Embeddings